

外国語

えいご
英語

問題1. 次の英文を読んで、各問いに答えなさい。

Problem 1. Read the following English writing and answer the questions.

There has never been anything quite like information technology before, but there have certainly been other complex technologies that needed simplifying. Joe Corn, a history professor at Stanford University, believes that the first example of a complex consumer technology was clocks, which arrived in the 1820s. Clocks were sold with user manuals, which featured entries such as “How to erect and regulate your device”. ①When sewing machines appeared in the 1840s, they came with 40-page manuals full of detailed instructions. Discouragingly, it took two generations until a trade publication was able to declare in the 1880s that “every woman now knows how to use one.”

At about the same time, the increase in technological complexity gathered pace. With electricity came new appliances, such as the phonograph, invented in 1877 by Thomas Alva Edison. According to Mr Norman, the computer-design guru, despite Mr Edison's genius for engineering he was a marketing moron, and his first phonograph was all but unusable (in fact, initially he had no particular uses in mind for it). For decades, Mr Edison fiddled with his technology, always going for the most impressive engineering solution. For instance, he chose cylinders over discs as the recording medium. It took a generation and the entry of a new rival, Emile Berliner, to prepare the phonograph for the mass market by making it easier to use (introducing discs instead of cylinders) and giving it a purpose (playing music). ②Mr Edison's companies foundered whereas Mr Berliner's thrived, and phonographs became ubiquitous, first as “gramophones” or “Victrolas”, the name of Mr Berliner's model, and ultimately as “record players”.

Another complex technology, with an even bigger impact, was the car. The first cars, in the early 1900s, were “mostly a burden and a challenge”, says Mr Corn. Driving one required skill in lubricating various moving parts, sending oil manually to the transmission, adjusting the spark plug, setting the choke, opening the throttle, wielding the crank and knowing what to do when the car broke down, which it invariably did. People at the time hired chauffeurs, says Mr Corn, mostly because they needed to have a mechanic at hand to fix the car, ③just as firms today need IT staff and households need teenagers to sort out their computers.

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平成17年度2月期 筑波大学大学院システム情報工学研究科コンピュータサイエンス専攻
入学試験問題

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問（1）下線部①と②を和訳せよ。

Question (1) Translate the underlined parts ① and ② into Japanese.

問（2）下線部③を日本語又は英語で適宜わかりやすく説明せよ。

Question (2) Explain the underline part ③ in detail in Japanese or English.

問（3）エジソンの蓄音機はなぜ商業的に成功しなかったのか、日本語又は英語で簡潔に説明せよ。

Question (3) Briefly explain why Mr Edison's phonograph did not succeed commercially in Japanese or English.

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問題2. 次の5つのコンピュータサイエンスの専門語から3つ選び、英語で簡単に説明しなさい。

Problem 2. Choose three out of the following five computer science technical terms, and make a brief explanation for them in English.

- (1) スタック (stacks)
- (2) 再帰呼出し (recursive calls)
- (3) オペレーティングシステム (operating systems)
- (4) オブジェクト指向 (object-oriented)
- (5) 認証 (authentication)